



MINING DEALER BEST PRACTICE

HMS 6030 Engine and Pump Drive PCR Stand

Maintenance and Repair

Application

Component Life Management

Component Renewal (CRC)

MARC Management

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November 2023
1.01-231107-1

1.0 Introduction

Planned component replacement is an event which occurs several times over the life of a mining machine such as the 6030 or similarly sized mining shovels. These machines seldom enter a workshop environment; hence the work is very demanding on those executing the shut-down. There are multiple sources of contamination and various inefficiencies associated with this type of work.

2.0 Best Practice Description

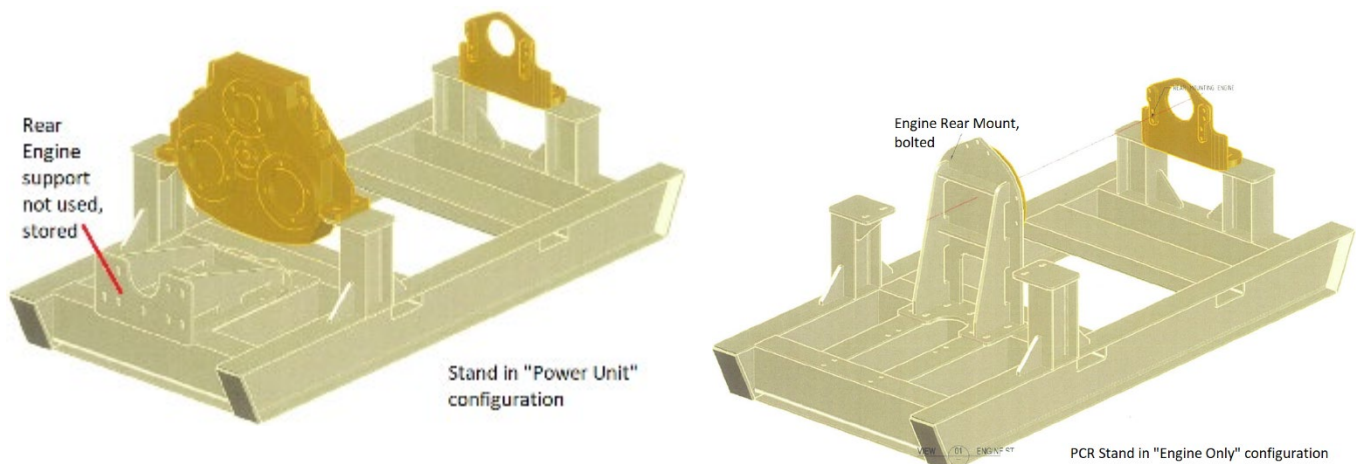
Supporting a fleet of 10 x 6030 with similar hours on a MARC, our dealership looked at options to take some of the work from the field into the workshop and reduce the time required for the planned replacement of engine, pump drive and pumps. Our maintenance strategy has these components, as well as some hoses around them, listed for replacement at roughly the same time.

We designed special stands which will allow us to join engine and pump drive (with coupling) in the workshop and then add the remaining power train / pumps onto the PTO. Through spare harnesses and new hoses, we can complete most of the power pack in the workshop and ship it to site as one unit, before the machine has even been parked up.

The removed power pack can then be shipped back to the workshop / branch and dismantled for overhaul in the workshop.

3.0 Implementation Steps

Key measurements of engine and pump drive were taken and a stand for each side power unit designed. This stand can also be used to ship just the engine on it's own, should this be required.



A few weeks before the machine is due to shut down, the spare engines and pump drives are installed to the stands. The team then proceed to install new/reman pumps into the position where pumps are due for replacement. In some instances, a field swap may still be necessary, e.g., if pump #3 has not yet achieved life due to an earlier replacement, the workshop power unit will not receive a new pump in this place.

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Once pumps have been installed the team can proceed installing harnesses and hoses. Once all is complete the power unit is adequately packaged / covered and prepared for shipping.



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On site, during the PCR shutdown, the old Left-hand and Right-hand power units are removed within 1 shift. After some clean-up and rubber mount replacement, the new power unit can be lifted into place and torqued down. All harness and hose connections can then be made.



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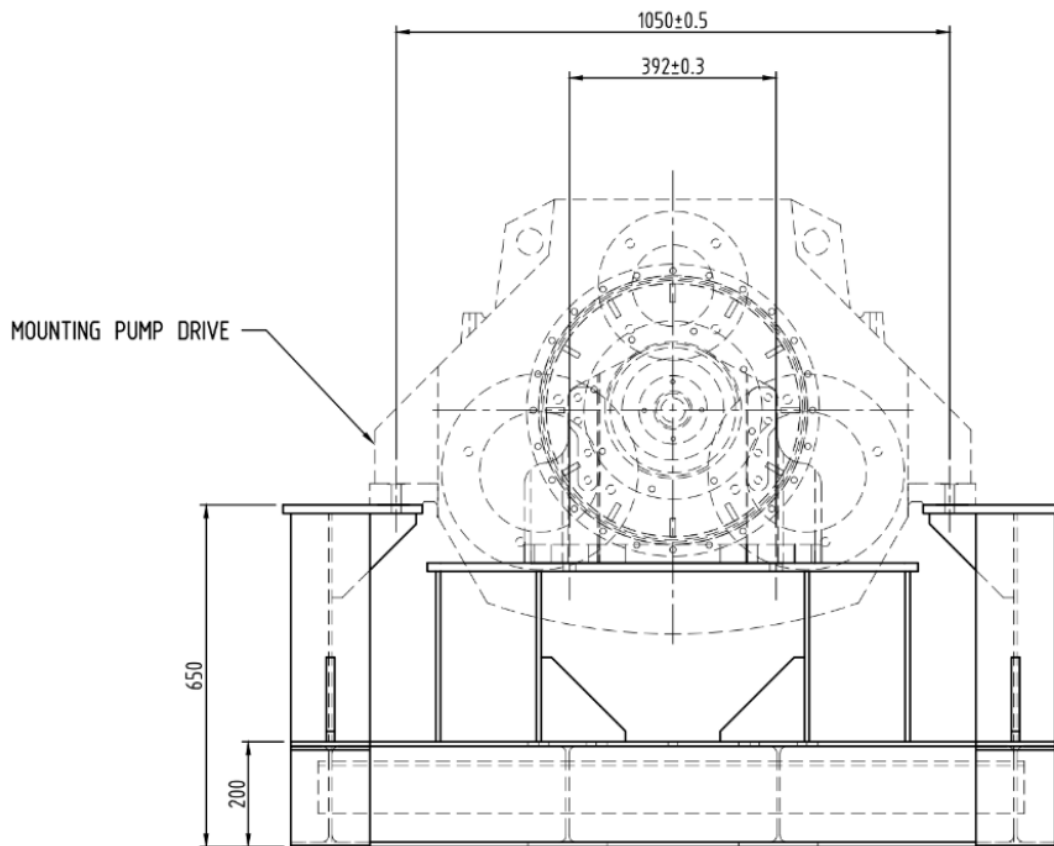
4.0 Benefits

- Time savings – less downtime for machine, ~24 hours per engine and pump combination
- Critical processes done in workshop.
- Better contamination control
- Shop labor is cheaper.
- Timing – build up of the power unit can be done in times of lower priority activities.
- Less waste – 1-time packaging only

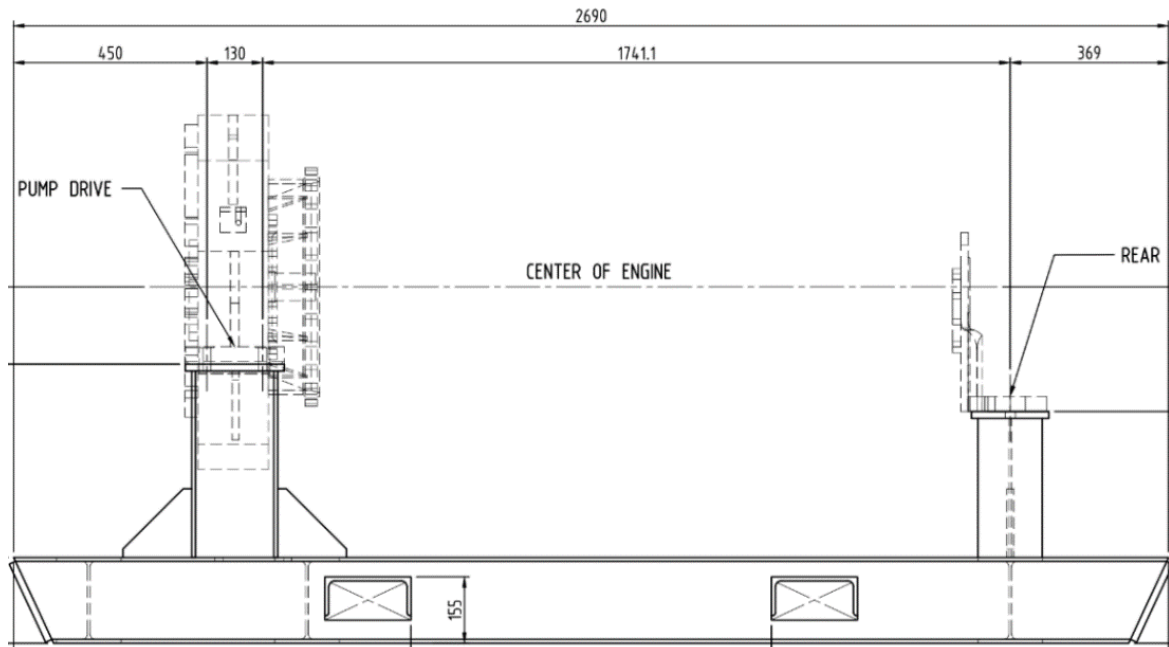
5.0 Resources Required

- Spare components and harnesses for the exchange
- Hose supply / hose shop
- Stands – cost around USD \$2,500, per stand.
- Overhead crane
- Suitable shop transport / large forklift
- Standard hand tools

6.0 Supporting Attachments / References



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- Drawings can be made available upon request (see Trakindo contact details below)

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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